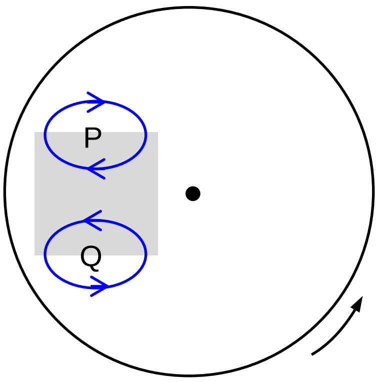
5 **(a)** The sections of the disc moving towards or away from the magnets experience a change in magnetic flux linkage. There will be induced e.m.f. and induced currents in the conducting disc.

By Lenz's law, the direction of the induced currents in the disc will oppose the change in the magnetic flux linkage. This produces a force that opposes the direction of the spin of the disc, causing it to slow down.

In slowing down, the mechanical / kinetic energy of the disc is converted to electrical energy as induced currents and eventually converted to thermal energy which is dissipated to the surroundings. This obeys the law of conservation of energy.

(b)



*At least one closed loop drawn at each region P and Q.
*Correct direction of eddy current for each loop.
*For each loop, part of loop in the magnetic field, part of it outside the magnetic field.

(c) As the disc is slowing down, its angular velocity is decreasing. This leads to a rate of change of magnetic flux linkage that is decreasing which produces a decreasing induced e.m.f.

The decreasing induced current produces an opposing / decelerating force that is decreasing in magnitude.

This causes the angular velocity of the disc to decrease at a decreasing rate. Hence its decrease is not linear (does not decrease at a constant rate).

(Angular deceleration is decreasing in magnitude.)

6 **(a)** The root mean square value of an alternating current is that value of the direct current